Seunghwan Um

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Research Statement

I'm a PhD student at Robotics Innovatory, Sungkyunkwan University, South Korea, under the supervision of **Prof. Hyouk Ryeol Choi**.
My research focuses on ***contact-rich robotic manipulation***, connecting physically intelligent gripper design, compliant control, and robot learning. My recent work combines fingertip force sensing, bilateral force-feedback teleoperation, imitation and reinforcement learning for robotic hands, and policy improvement through human corrections.

Education

- **Sungkyunkwan University (SKKU)** — *Mar. 2022 - Present*
  Ph.D in Mechanical Engineering — Suwon, Republic of Korea
  - Expected graduation: February 2027. Dissertation: *Compliant Mechanisms and Systems for Contact-Rich Robotic Manipulation*.
- **Hanyang University (ERICA)** — *Mar. 2018 - Feb. 2022*
  B.S in Mechanical Engineering, Degree Honors: Cum Laude — Ansan, Republic of Korea

Publications

J=Journal, C=Conference, S=In Submission, P=Patent, T=Thesis

[J.6] **PALM-Gripper: Integrated Gripper with Parallel Adaptable Mechanism for Shelf Picking in Logistics**
Seunghwan Um, Yeong Gwang Son, Juyong Hong, Chun Soo Kim, et al. and Hyouk Ryeol Choi*.
*IEEE/ASME Transactions on Mechatronics (**TMECH**), 2026.*

[J.5] **Toward Reliable Bin-Picking: Collision-Aware Robotic Design and Control Strategy for Heavily Cluttered Environment**
Seunghwan Um, Yeong Gwang Son, Jaeyun Sim, Hyouk Ryeol Choi*
*IEEE Robotics and Automation Practice (**RA-P**), 2026, The paper was invited for publication.*

[J.4] **Plug-and-Play Shape Matching Module for Zero-Shot Mesh-Free Grasp Refinement on Unknown Objects**
Juyong Hong, Yeong Gwang Son, Seunghwan Um, Hyouk Ryeol Choi*.
*IEEE Robotics and Automation Letters (**RA-L**), 2025. Presented at ICRA 2026.*

[J.3] **Corner-Grasp: Multi-Action Grasp Detection and Active Gripper Adaptation for Grasping in Cluttered Environments**
Yeong Gwang Son, Seunghwan Um, Juyong Hong, Tat Hieu Bui, Hyouk Ryeol Choi*.
ArXiv, 2025. *Under Revision at Autonomous Robots.*

[J.2] **Development of Adaptive Gripper Enhancing Power Grasp Range and Linearity**
Issac Rhee, Chun Soo Kim, Heeyeon Jeong, Seunghwan Um, and Hyouk Ryeol Choi* et al.
IEEE Access, 2024.

[J.1] **ReC-Gripper: A Reconfigurable Combined Suction and Fingered Gripper for Various Logistics Picking and Stowing Tasks**
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Hyouk Ryeol Choi*
*IEEE Robotics and Automation Letters (**RA-L**), Presented in ICRA 2024.*

[C.1] **Overcoming Heavy Clutter: Utilizing the Hybrid Grasping Network and Gripper**
Seunghwan Um, Yeong Gwang Son, Tat Hieu Bui, Ho Sang Jung, and Hyouk Ryeol Choi*
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024*
Workshop: Benchmarking via Competitions in Robotic Grasping and Manipulation ***[Best Extended Abstract]***

[S.3] **Mesh-Free Dexterous Robot Demonstrations from Human RGB-D Video and Teleoperation**
Juyong Hong, Seunghwan Um, Hyouk Ryeol Choi*.
*Submitted to **IEEE Access**, Under Review.*

[S.2] **Pointwise Uncertainty-Aware Fusion of LiDAR and Radar for Dynamic Occupancy Grid Mapping and Navigation**
Hyunchang Kang, Minh Thang Bui, Sung-Jin Han, Youngbin Song, Jaeyun Sim, Seunghwan Um, Hyouk Ryeol Choi*.
*Submitted to IEEE Robotics and Automation Letters (**RA-L**), Under Review.*

[S.1] **Development of an Ultra-High Holding Force Density Halbach-Electro-Permanent Magnet Gripper for Robotic Applications**
Heeyeon Jeong, Seunghwan Um, Hyunchang Kang, Jaeyun Sim, Hyouk Ryeol Choi*.
*Submitted to IEEE Robotics and Automation Letters (**RA-L**), Under Review.*

Patents

[P.6] **Horizontal Adaptive Hybrid Gripper and Robot Including the Same**.
Seunghwan Um, Chun Soo Kim, Issac Rhee, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2024-0131112. Application Date: 2024.09.26.

[P.5] **Gripper Including Multi-Stage Translational Mechanism**.
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2023-0125840. Application Date: 2023.09.20.

[P.4] **Gripper Including Telescopic Mechanism**.
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Application No. 10-2023-0124012. Application Date: 2023.09.18.

[P.3] **Suction Gripper Capable of Translational and Rotational Movements**.
Chun Soo Kim, Issac Rhee, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0093340. Registration Date: 2023.10.18.

[P.2] **Adaptive Gripper Capable of Parallel Motion**.
Issac Rhee, Chun Soo Kim, Seunghwan Um, Heeyeon Jeong, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0077512. Registration Date: 2023.12.06.

[P.1] **Hybrid Gripper Capable of Bin Picking and Shelf Picking**.
Seunghwan Um, Heeyeon Jeong, Chun Soo Kim, Issac Rhee, and Yoon Haeng Lee.
Korean Intellectual Property Office, Patent No. 10-2023-0076248. Publication Date: 2024.12.23.

Honors and Awards

- **KSME National Student Design Competition – Prize Winner** *2026*
  Handling Hand team; final ranking to be announced [Handling Hand]
- **Robot Engineering and Design Show (RED Show)** *Feb. 2026*
  Korea Robotics Society Annual Conference (KRoC) [KRoC RED Show]
- **Samsung Humantech Paper Award** *Feb. 2025*
  Samsung Electronics Co., Ltd. [Samsung Humantech]
- **9th Robotic Grasping of Manipulation Competition - Picking in Clutter** *May. 2024*
  IEEE, IEEE RAS [RGMC 2024]
- **KSME Student Creative Design Competition** *Oct. 2021*
  The Korean Society of Mechanical Engineers (KSME) [YouTube]
- **Creative and Intelligent Robot Contest** *Sep. 2021*
  Daejeon Metropolitan City, Chungnam National University [YouTube]

Experience

- **CarbonSix** *Oct. 2024 - Nov. 2024, Jun. 2026 - Jul. 2026*
  Researcher Gangnam-daero, Seocho-gu, Seoul, Republic of Korea
  ◦ Designed direct teaching device with gravity compensation function, and camera calibration device.
- **PRIMETECH** *May 2026 - Aug. 2026*
  Perception Development (Contract) Asan-si, Chungcheongnam-do, Republic of Korea
  ◦ Developed Windows-based RGB-D perception for Samsung Mobile's Vietnam factory, including pallet/box inspection, empty-slot detection, and box-pose and QR-face orientation estimation.
- **AIDIN ROBOTICS - Cobot Solution Team** *May 2022 - Present*
  Researcher Anyang-si, Gyeonggi-do, Republic of Korea
  ◦ Designed grippers for shelf-picking solutions, contributing to efficient logistics automation.
- **Korea Institute of Industrial Technology (KITECH)** *Oct 2021 - Dec 2021*
  Research Student Ansan-si, Gyeonggi-do, Republic of Korea
  ◦ Designed and developed control systems for a 2-DoF manipulator, enhancing its performance and accuracy.
- **Wall Climbing Car (WCC) [Undergraduate Project]** *Dec 2020 - Oct 2021*
  Team Leader Ansan-si, Gyeonggi-do, Republic of Korea
  ◦ Designed a wall-climbing car's propeller frame and control system.

PERSONAL PROJECTS

- **Learning from Human Corrections | Residual Learning** *2026*
  [🌐] [GitHub]
  - Trained a residual policy from human corrections to refine a frozen diffusion-based base policy for contact-rich robotic box insertion.
  - Evaluated at the Original position and an unseen backward target shift of 3 cm; the base policy used absolute-action demonstrations collected only at the Original position.
- **Scooping Grasping with Robotic Hand | Fingertip Force Sensing, Diffusion Policy** *May 2026 - Present*
  [🌐] [GitHub]
  - Enabled a rigid-fingertip robotic hand to scoop and lift objects through compliant contact with the supporting surface.
  - Transformed fingertip sensor wrenches into the wrist frame and combined them with wrist F/T feedback for compliant manipulator control during scooping.
  - Implemented compliance control on the Doosan M0609 with a momentum-based disturbance observer (De Luca formulation) for external-contact estimation, together with gravity, tool, and friction compensation.
  - Demonstrated diffusion policy rollouts grasping objects of various sizes and collected grasping demonstrations for policy learning.
- **Robotic-Hand Grasping | Reinforcement Learning** *2026*
  AIDIN robotic hand and Doosan M0609 [🌐]
  - Trained grasp-and-lift policies in Isaac Sim with 131-dimensional observations and 21 actions controlling 6 arm and 15 hand joints.
  - Integrated D455, SAM3, and FoundationPose object-pose estimates with robot joint feedback for hardware policy execution; applied an initial policy on the real robot.
  - Evaluated later checkpoints in simulation across four object families and studied hand reachability and contact pads in MuJoCo. Repeated hardware success-rate evaluation is pending.
- **Bilateral Force-Feedback Teleoperation for Dual Arms and Robotic Hands** *2026*
  [🌐]
  - Developed gravity-compensated leader-arm torque control and Jacobian-transpose force reflection for bilateral arm–hand teleoperation.
  - Presented the system at a KRoC 2026 special session; the related team project received 1st place at the KRoC 2026 RED Show.
- **Impedance Control with Doosan M0609** *Jan. 2026 - Present*
  [🌐]
  - Implemented a momentum-based disturbance observer with gravity, tool, and friction compensation for compliant contact.
  - Demonstrated contact maintenance under continuous forces and impacts, and pivoting in constrained environments.
- **Admittance Control with Rainbow Robotics RB10**
  [🌐]
  - Applied force-feedback admittance control to pushing and grasping objects in cluttered environments.
- **Handling Hand | Low-Cost Modular 16-DoF Robotic Hand** *2026*
  Collaborative project with Jeonghwan Park and Taehwan Yoon; advisor: Hyouk Ryeol Choi [🌐]
  - Participated in a modular robotic-hand project using commercial servo components rearranged inside the fingers.
  - The team demonstrated physical grasping and manipulation and used MuJoCo to assess the hand design.

PROJECTS

- **Development of a K-Logistics Humanoid Robot Integrated with a High-Sensitivity Robotic Hand Based on a Multimodal AI Foundation Model** *Sep 2025 - Present*
  *Participating organizations: AIDIN ROBOTICS, **SKKU**, KETI, CJ Logistics* Republic of Korea
  - **Research Objective**: Automation of contact-rich manipulation tasks using a humanoid robot equipped with a high-sensitivity robotic hand.
  - Development of a teleoperation system for constructing multimodal datasets incorporating force information
  - Design of force-control strategies and learning-based policies capable of handling contact-rich interactions
- **Development of Smart Vision System and All-in-One Universal Gripper for Multi-Various Random Piece Picking** *May 2022 - 2024*
  *Participating organizations: **SKKU**, AIDIN ROBOTICS, KITECH, CJ Logistics* Republic of Korea
  - **Research Objective**: Developing robotic picking system including a gripper and vision system for piece picking in a logistics environment.
  - Designing an integrated gripper capable of various grasping strategy for a shelf environment among logistics environments
  - Researching grasping strategy that can pick objects while avoiding external constraints in a shelf environment

GRANT

- **BK21 Research Encouragement Scholarship** *Fall. 2024, and Spring. 2026*
- **Teaching Assistantship Scholarship** *Spring. 2022, Fall. 2022*
- **Brain Hanyang, Academic Excellence Scholarship** *Spring. 2018*

REVIEW SERVICE

- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**
- **Scientific Reports**

ADVISING EXPERIENCE

- **Research Mentoring**
  Proposed experimental methodology for a journal publication
  - Proposed the experimental design for the T-MECH paper *Design and Control of 2-DOF Differential Joint Driven by Soft Pneumatic Actuator for Versatile Soft Robotic Manipulator*; due to project circumstances, I was unable to be included as a co-author.
- **Undergraduate Student Mentoring**
  Mentored undergraduate students, leading to awards in university-level and national competitions
  - Guided undergraduate students in robotics research and project development, resulting in awards at both university-level and national competitions.
  - Mentees: Taehwan Yoon, Jeonghwan Park, Sihun Ryu, Hyunjun Jo, Kihyeok Kwon, Seongmin Kim, Seongmin Ryu.